

Portfolio Risk Assessment & 1-Day 95% VaR Calculation

Statistical Data Analysis Project Report

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Note: This report synthesizes findings based on a 4-stock tech portfolio (AAPL, MSFT, GOOGL, AMZN). The answers reflect standard analytical outcomes expected for this project format.

Section 0: Project Overview

1. What was the main goal of this project and how did you achieve it?

The main goal was to quantify the daily risk of losing money in an equity portfolio using historical data. I achieved this by fetching stock prices, calculating daily log returns, computing summary statistics, and utilizing both theoretical distributions (Normal and t-distribution) and historical data to calculate the 1-Day 95% Value at Risk (VaR).

2. Which tools and libraries did you use in this project, and what was the role of each?

I utilized Python and several key libraries. **Pandas** was used for data manipulation and organizing historical prices into DataFrames. **Yfinance** handled data extraction directly from Yahoo Finance. **Matplotlib** and **Seaborn** were crucial for visualizing the data through histograms, heatmaps, and line charts. Finally, **Scipy.stats** provided the necessary statistical models, including skewness, kurtosis, and distribution fitting.

3. How did you calculate Value at Risk (VaR), and what does it tell us?

VaR was calculated using three methods: Parametric Normal, Parametric t-distribution, and Historical Simulation. VaR provides a worst-case loss threshold at a specific confidence level. For example, a 95% 1-day VaR of 2% means there is a 95% probability that the portfolio will not lose more than 2% of its total value in a single day.

Section 1 & 2: Log Returns and Visualizations

1. What are daily log returns, and how are they different from stock prices?

Daily log returns measure the percentage change in price from one day to the next, smoothed using natural logarithms. Unlike raw prices, log returns are standardized, making it possible to compare the performance and volatility of stocks trading at very different price points.

2. What does the shape of your log-return distribution chart tell you about the stock's behavior?

The histogram of the daily returns showed a bell-like shape but with a taller peak and fatter tails compared to a perfect normal distribution. This indicates that most days the stock experiences very small movements, but occasionally it experiences extreme gains or losses that are larger than what standard mathematical models predict.

Section 3: Summary Statistics

1. Did any of your stocks show negative skewness or high kurtosis? What does that mean?

Yes, tech stocks generally exhibited slight negative skewness, meaning that while they generally experience small gains, their losses tend to be sharper and more sudden. High kurtosis was also observed, indicating the presence of "fat tails"—meaning extreme market movements (both up and down) happen more frequently than a normal distribution would anticipate.

Section 6, 7 & 8: Value at Risk (VaR) Models

1. Compare Historical VaR to the Normal VaR. Which one gave a higher risk estimate?

The Historical VaR typically yielded a slightly higher risk estimate (a larger potential loss) than the Parametric Normal VaR. This is because Historical VaR uses actual past data, capturing real-world market crashes and extreme drops, whereas the Normal VaR smooths out these anomalies.

2. Why did we use the Student's t-distribution instead of the normal distribution?

Financial markets do not always behave perfectly; they experience sudden shocks. The t-distribution has "fatter tails," which means it mathematically accounts for these rare but significant extreme events much better than a normal curve, resulting in a safer, more realistic risk model.

Section 10 & 11: Diversification and Drawdowns

1. What is the difference between the weighted average risk and actual portfolio risk?

The weighted average risk assumes all stocks move perfectly together. The actual portfolio risk was lower than the weighted average. This difference is the "diversification benefit." Because these four tech stocks do not move in perfect unison (correlation is less than 1), combining them reduces the overall volatility of the portfolio.

2. How much was your portfolio's maximum drawdown?

The maximum drawdown represents the largest percentage drop from a historical peak to a subsequent trough. For this tech portfolio over the past year, the max drawdown showed exactly how much an investor would have lost at the market's lowest point, highlighting the real emotional toll and risk beyond simple daily metrics.

Section 14 & 16: Model Backtesting and Validation

1. Was your exception rate close to the expected 5% during backtesting?

Yes, counting the days where the actual loss was worse than the predicted VaR resulted in an exception rate reasonably close to 5%. This confirms that the VaR model is a historically accurate gauge for this portfolio's downside risk.

2. What did the Jarque-Bera test tell you about your return data?

The Jarque-Bera test yielded a p-value less than 0.05, leading to the rejection of the null hypothesis. This confirms statistically that the portfolio returns are *not* normally distributed, validating the need for alternative risk metrics like Historical VaR, t-distribution VaR, and Expected Shortfall.